Supplementary Material

A Family of Classifying Criteria Emerging from Least-Squares Matching Between Partition Incidence Matrices

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Results of Computational Experiments

Synthetic data

Synthetic datasets used in the experiments.

Dataset	V	K	α
1	6	4	0.5
2	6	4	0.85
3	15	4	0.5
4	15	4	0.85
5	6	8	0.5
6	6	8	0.85
7	15	8	0.5
8	15	8	0.85
9	6	15	0.5
10	6	15	0.85
11	15	15	0.5
12	15	15	0.85

Table 1.1: $N = 2000$, $V = 6$, $K = 4$, $\alpha = 0.5$, $n_{min} = 50$, tree_depth = 3.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	0.9915	0.9933	0.9933	**0.9936**	0.9934	0.9929	0.9935	0.9934	**0.9936**
	Std	0.0038	0.0030	0.0030	0.0031	0.0030	0.0033	0.0028	0.0029	0.0031
Precision	Mean	0.9916	0.9934	0.9934	**0.9937**	0.9935	0.9930	0.9935	0.9935	0.9936
	Std	0.0038	0.0029	0.0029	0.0031	0.0030	0.0032	0.0028	0.0028	0.0031
Recall	Mean	0.9915	0.9933	0.9933	**0.9936**	0.9934	0.9929	0.9935	0.9934	**0.9936**
	Std	0.0038	0.0030	0.0030	0.0031	0.0030	0.0033	0.0028	0.0029	0.0031
F1 score	Mean	0.9915	0.9933	0.9933	**0.9936**	0.9934	0.9929	0.9935	0.9934	**0.9936**
	Std	0.0038	0.0030	0.0030	0.0031	0.0030	0.0033	0.0028	0.0029	0.0031
ARI	Mean	0.9775	0.9822	0.9822	**0.9831**	0.9824	0.9812	0.9827	0.9824	0.9830
	Std	0.0099	0.0077	0.0077	0.0081	0.0079	0.0086	0.0072	0.0076	0.0080

Table 1.2: $N = 2000$, $V = 6$, $K = 4$, $\alpha = 0.5$, $n_{min} = 50$, tree_depth = 4.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	0.9926	0.9932	0.9931	0.9933	**0.9935**	**0.9935**	0.9931	0.9933	0.9932
	Std	0.0040	0.0038	0.0037	0.0037	0.0033	0.0035	0.0034	0.0036	0.0037
Precision	Mean	0.9927	0.9933	0.9932	0.9934	**0.9936**	0.9935	0.9931	0.9933	0.9932
	Std	0.0040	0.0038	0.0036	0.0036	0.0033	0.0034	0.0033	0.0036	0.0037
Recall	Mean	0.9926	0.9932	0.9931	0.9933	**0.9935**	**0.9935**	0.9931	0.9933	0.9932
	Std	0.0040	0.0038	0.0037	0.0037	0.0033	0.0035	0.0034	0.0036	0.0037
F1 score	Mean	0.9926	0.9932	0.9931	0.9933	**0.9935**	**0.9935**	0.9931	0.9933	0.9932
	Std	0.0040	0.0038	0.0037	0.0037	0.0033	0.0035	0.0034	0.0036	0.0037
ARI	Mean	0.9804	0.9821	0.9817	0.9823	**0.9828**	0.9827	0.9817	0.9821	0.9819
	Std	0.0104	0.0100	0.0095	0.0095	0.0087	0.0090	0.0086	0.0094	0.0096

Table 2.1: $N = 2000$, $V = 6$, $K = 4$, $\alpha = 0.85$, $n_{min} = 50$, tree_depth = 3.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	0.7194	0.7312	0.7254	**0.7356**	0.7245	0.7245	0.7186	0.7271	0.7334
	Std	0.0216	0.0196	0.0218	0.0182	0.0212	0.0183	0.0250	0.0223	0.0183
Precision	Mean	0.7432	0.7408	0.7400	**0.7440**	0.7381	0.7388	0.7146	0.7386	0.7401
	Std	0.0188	0.0184	0.0188	0.0176	0.0203	0.0207	0.0342	0.0191	0.0178
Recall	Mean	0.7194	0.7312	0.7254	**0.7356**	0.7245	0.7245	0.7186	0.7271	0.7334
	Std	0.0216	0.0196	0.0218	0.0182	0.0212	0.0183	0.0250	0.0223	0.0183
F1 score	Mean	0.7205	0.7300	0.7259	**0.7337**	0.7246	0.7248	0.7048	0.7264	0.7300
	Std	0.0215	0.0191	0.0198	0.0179	0.0200	0.0182	0.0353	0.0206	0.0185
ARI	Mean	0.4245	0.4544	0.4393	**0.4642**	0.4383	0.4362	0.4635	0.4459	0.4629
	Std	0.0383	0.0377	0.0427	0.0332	0.0423	0.0376	0.0308	0.0436	0.0341

Table 2.2: $N = 2000$, $V = 6$, $K = 4$, $\alpha = 0.85$, $n_{min} = 50$, tree_depth = 4.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	0.7451	**0.7576**	0.7529	0.7567	0.7518	0.7493	0.7446	0.7517	0.7529
	Std	0.0166	0.0160	0.0173	0.0161	0.0185	0.0187	0.0171	0.0161	0.0160
Precision	Mean	0.7567	**0.7628**	0.7616	0.7608	0.7627	0.7617	0.7565	0.7565	0.7566
	Std	0.0187	0.0175	0.0196	0.0175	0.0207	0.0204	0.0201	0.0181	0.0170
Recall	Mean	0.7451	**0.7576**	0.7529	0.7567	0.7518	0.7493	0.7446	0.7517	0.7529
	Std	0.0166	0.0160	0.0173	0.0161	0.0185	0.0187	0.0171	0.0161	0.0160
F1 score	Mean	0.7429	**0.7571**	0.7533	0.7553	0.7529	0.7508	0.7448	0.7503	0.7507
	Std	0.0181	0.0164	0.0173	0.0163	0.0180	0.0183	0.0173	0.0172	0.0175
ARI	Mean	0.4753	0.4937	0.4810	**0.4960**	0.4769	0.4707	0.4798	0.4838	0.4894
	Std	0.0331	0.0273	0.0316	0.0286	0.0337	0.0356	0.0300	0.0287	0.0268

Table 3.1: $N = 2000$, $V = 15$, $K = 4$, $\alpha = 0.5$, $n_{min} = 50$, tree_depth $= 3$.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	**0.9994**	0.9977	0.9975	0.9978	0.9974	0.9975	0.9982	0.9974	0.9978
	Std	0.0010	0.0025	0.0026	0.0024	0.0025	0.0024	0.0022	0.0026	0.0024
Precision	Mean	**0.9994**	0.9977	0.9975	0.9978	0.9975	0.9975	0.9983	0.9974	0.9978
	Std	0.0010	0.0025	0.0026	0.0024	0.0025	0.0024	0.0022	0.0025	0.0024
Recall	Mean	**0.9994**	0.9977	0.9975	0.9978	0.9974	0.9975	0.9982	0.9974	0.9978
	Std	0.0010	0.0025	0.0026	0.0024	0.0025	0.0024	0.0022	0.0026	0.0024
F1 score	Mean	**0.9994**	0.9977	0.9975	0.9978	0.9974	0.9975	0.9982	0.9974	0.9978
	Std	0.0010	0.0025	0.0026	0.0024	0.0025	0.0024	0.0022	0.0026	0.0024
ARI	Mean	**0.9985**	0.9939	0.9934	0.9941	0.9933	0.9935	0.9954	0.9930	0.9941
	Std	0.0027	0.0065	0.0070	0.0064	0.0066	0.0064	0.0059	0.0068	0.0063

Table 3.2: $N = 2000$, $V = 15$, $K = 4$, $\alpha = 0.5$, $n_{min} = 50$, tree_depth $= 4$.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	**0.9994**	0.9977	0.9975	0.9978	0.9975	0.9976	0.9984	0.9974	0.9978
	Std	0.0010	0.0025	0.0026	0.0024	0.0025	0.0024	0.0023	0.0026	0.0024
Precision	Mean	**0.9994**	0.9977	0.9975	0.9978	0.9975	0.9976	0.9984	0.9974	0.9978
	Std	0.0010	0.0025	0.0025	0.0024	0.0024	0.0023	0.0022	0.0025	0.0024
Recall	Mean	**0.9994**	0.9977	0.9975	0.9978	0.9975	0.9976	0.9984	0.9974	0.9978
	Std	0.0010	0.0025	0.0026	0.0024	0.0025	0.0024	0.0023	0.0026	0.0024
F1 score	Mean	**0.9994**	0.9977	0.9975	0.9978	0.9975	0.9976	0.9984	0.9974	0.9978
	Std	0.0010	0.0025	0.0026	0.0024	0.0025	0.0024	0.0023	0.0026	0.0024
ARI	Mean	**0.9985**	0.9939	0.9935	0.9941	0.9935	0.9937	0.9957	0.9930	0.9941
	Std	0.0027	0.0065	0.0068	0.0064	0.0065	0.0063	0.0059	0.0068	0.0063

Table 4.1: $N = 2000$, $V = 15$, $K = 4$, $\alpha = 0.85$, $n_{min} = 50$, tree_depth $= 3$.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	0.9515	0.9514	0.9502	**0.9521**	0.9490	0.9464	0.9428	0.9498	0.9509
	Std	0.0068	0.0076	0.0088	0.0079	0.0102	0.0147	0.0311	0.0085	0.0082
Precision	Mean	0.9527	0.9525	0.9514	**0.9532**	0.9502	0.9477	0.9466	0.9512	0.9521
	Std	0.0065	0.0074	0.0085	0.0076	0.0101	0.0143	0.0167	0.0080	0.0077
Recall	Mean	0.9515	0.9514	0.9502	**0.9521**	0.9490	0.9464	0.9428	0.9498	0.9509
	Std	0.0068	0.0076	0.0088	0.0079	0.0102	0.0147	0.0311	0.0085	0.0082
F1 score	Mean	0.9515	0.9515	0.9502	**0.9522**	0.9490	0.9464	0.9416	0.9499	0.9510
	Std	0.0068	0.0076	0.0088	0.0079	0.0102	0.0147	0.0405	0.0084	0.0081
ARI	Mean	0.8746	0.8744	0.8715	**0.8763**	0.8686	0.8621	0.8582	0.8700	0.8726
	Std	0.0168	0.0189	0.0219	0.0196	0.0256	0.0367	0.0417	0.0212	0.0205

Table 4.2: $N = 2000$, $V = 15$, $K = 4$, $\alpha = 0.85$, $n_{min} = 50$, tree_depth = 4.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	**0.9660**	0.9640	0.9627	0.9655	0.9627	0.9618	0.9572	0.9608	0.9608
	Std	0.0078	0.0084	0.0087	0.0089	0.0089	0.0092	0.0087	0.0083	0.0085
Precision	Mean	**0.9666**	0.9646	0.9634	0.9661	0.9633	0.9624	0.9582	0.9616	0.9615
	Std	0.0076	0.0082	0.0085	0.0087	0.0088	0.0091	0.0084	0.0083	0.0080
Recall	Mean	**0.9660**	0.9640	0.9627	0.9655	0.9627	0.9618	0.9572	0.9608	0.9608
	Std	0.0078	0.0084	0.0087	0.0089	0.0089	0.0092	0.0087	0.0083	0.0085
F1 score	Mean	**0.9660**	0.9640	0.9628	0.9656	0.9627	0.9618	0.9573	0.9609	0.9608
	Std	0.0078	0.0083	0.0087	0.0089	0.0089	0.0092	0.0087	0.0083	0.0084
ARI	Mean	**0.9113**	0.9061	0.9029	0.9101	0.9029	0.9006	0.8892	0.8980	0.8977
	Std	0.0199	0.0212	0.0221	0.0225	0.0229	0.0236	0.0219	0.0210	0.0218

Table 5.1: $N = 2000$, $V = 6$, $K = 8$, $\alpha = 0.5$, $n_{min} = 50$, tree_depth = 3.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	**0.8639**	0.6444	0.6156	0.8579	0.6025	0.5420	0.7398	0.6181	0.6470
	Std	0.0158	0.0730	0.0435	0.0334	0.0309	0.0616	0.0428	0.0435	0.0757
Precision	Mean	**0.8033**	0.5973	0.5711	0.7949	0.5722	0.5622	0.6743	0.5868	0.6136
	Std	0.0190	0.1088	0.0917	0.0477	0.0833	0.0843	0.0644	0.1007	0.1117
Recall	Mean	**0.8639**	0.6444	0.6156	0.8579	0.6025	0.5420	0.7398	0.6181	0.6470
	Std	0.0158	0.0730	0.0435	0.0334	0.0309	0.0616	0.0428	0.0435	0.0757
F1 score	Mean	**0.8236**	0.5575	0.5375	0.8160	0.5317	0.4821	0.6775	0.5362	0.5600
	Std	0.0188	0.0866	0.0490	0.0434	0.0328	0.0596	0.0491	0.0495	0.0896
ARI	Mean	**0.8466**	0.6129	0.5236	0.8406	0.4729	0.3596	0.6552	0.5399	0.6124
	Std	0.0181	0.0944	0.0951	0.0316	0.0717	0.0916	0.0639	0.0910	0.0924

Table 5.2: $N = 2000$, $V = 6$, $K = 8$, $\alpha = 0.5$, $n_{min} = 50$, tree_depth = 4.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	**0.9896**	0.8782	0.8020	0.9888	0.7602	0.6780	0.8769	0.8166	0.8800
	Std	0.0038	0.0676	0.0723	0.0049	0.0556	0.0634	0.0423	0.0703	0.0699
Precision	Mean	**0.9899**	0.8998	0.8498	0.9892	0.8239	0.7611	0.8395	0.8561	0.9052
	Std	0.0036	0.0874	0.0929	0.0047	0.0954	0.0963	0.0723	0.0945	0.0881
Recall	Mean	**0.9896**	0.8782	0.8020	0.9888	0.7602	0.6780	0.8769	0.8166	0.8800
	Std	0.0038	0.0676	0.0723	0.0049	0.0556	0.0634	0.0423	0.0703	0.0699
F1 score	Mean	**0.9896**	0.8486	0.7598	0.9888	0.7140	0.6300	0.8399	0.7748	0.8506
	Std	0.0038	0.0876	0.0856	0.0049	0.0624	0.0652	0.0548	0.0844	0.0905
ARI	Mean	**0.9765**	0.8556	0.7482	0.9745	0.6763	0.5446	0.8574	0.7719	0.8563
	Std	0.0083	0.0661	0.1088	0.0111	0.0981	0.1018	0.0450	0.1005	0.0688

Table 6.1: $N = 2000$, $V = 6$, $K = 8$, $\alpha = 0.85$, $n_{min} = 50$, tree_depth = 3.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	0.5947	**0.5988**	0.5896	0.5712	0.5791	0.5535	0.5055	0.5916	0.5970
	Std	0.0224	0.0216	0.0247	0.0357	0.0321	0.0316	0.0251	0.0218	0.0212
Precision	Mean	0.5402	**0.5420**	0.5345	0.5063	0.5258	0.4918	0.4137	0.5386	0.5405
	Std	0.0252	0.0205	0.0329	0.0571	0.0398	0.0475	0.0567	0.0272	0.0192
Recall	Mean	0.5947	**0.5988**	0.5896	0.5712	0.5791	0.5535	0.5055	0.5916	0.5970
	Std	0.0224	0.0216	0.0247	0.0357	0.0321	0.0316	0.0251	0.0218	0.0212
F1 score	Mean	0.5530	**0.5571**	0.5444	0.5145	0.5330	0.4958	0.4099	0.5474	0.5550
	Std	0.0235	0.0226	0.0299	0.0521	0.0361	0.0424	0.0344	0.0252	0.0221
ARI	Mean	0.3678	**0.3718**	0.3640	0.3701	0.3535	0.3283	0.3697	0.3655	0.3706
	Std	0.0229	0.0266	0.0270	0.0240	0.0340	0.0328	0.0232	0.0255	0.0258

Table 6.2: $N = 2000$, $V = 6$, $K = 8$, $\alpha = 0.85$, $n_{min} = 50$, tree_depth = 4.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	0.6434	0.6496	0.6495	0.6380	0.6453	0.6303	0.5874	**0.6512**	0.6510
	Std	0.0191	0.0199	0.0196	0.0309	0.0219	0.0262	0.0294	0.0191	0.0196
Precision	Mean	0.6599	0.6600	0.6619	0.6444	0.6574	0.6458	0.5783	**0.6626**	0.6606
	Std	0.0209	0.0218	0.0223	0.0394	0.0209	0.0255	0.0551	0.0222	0.0202
Recall	Mean	0.6434	0.6496	0.6495	0.6380	0.6453	0.6303	0.5874	**0.6512**	0.6510
	Std	0.0191	0.0199	0.0196	0.0309	0.0219	0.0262	0.0294	0.0191	0.0196
F1 score	Mean	0.6388	0.6439	0.6430	0.6297	0.6399	0.6263	0.5548	**0.6451**	0.6446
	Std	0.0194	0.0206	0.0194	0.0384	0.0212	0.0269	0.0419	0.0201	0.0204
ARI	Mean	0.3846	0.3923	0.3926	0.3869	0.3879	0.3709	0.3856	0.3940	**0.3945**
	Std	0.0246	0.0260	0.0279	0.0260	0.0295	0.0330	0.0271	0.0259	0.0265

Table 7.1: $N = 2000$, $V = 15$, $K = 8$, $\alpha = 0.5$, $n_{min} = 50$, tree_depth = 3.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	**0.9949**	0.6652	0.6709	0.6745	0.6762	0.6612	0.6432	0.6713	0.6630
	Std	0.0037	0.0804	0.0874	0.0712	0.0874	0.1001	0.0704	0.0873	0.0786
Precision	Mean	**0.9950**	0.5585	0.5851	0.5540	0.5945	0.6018	0.5276	0.5821	0.5530
	Std	0.0036	0.0861	0.0833	0.0797	0.0811	0.0829	0.0833	0.0845	0.0851
Recall	Mean	**0.9949**	0.6652	0.6709	0.6745	0.6762	0.6612	0.6432	0.6713	0.6630
	Std	0.0037	0.0804	0.0874	0.0712	0.0874	0.1001	0.0704	0.0873	0.0786
F1 score	Mean	**0.9949**	0.5821	0.5988	0.5867	0.6068	0.5942	0.5575	0.5974	0.5781
	Std	0.0037	0.0911	0.0961	0.0813	0.0955	0.1076	0.0814	0.0964	0.0895
ARI	Mean	**0.9885**	0.5971	0.5678	0.6338	0.5719	0.5494	0.5572	0.5748	0.6011
	Std	0.0084	0.1275	0.1297	0.1158	0.1258	0.1364	0.1113	0.1309	0.1248

Table 7.2: $N = 2000$, $V = 15$, $K = 8$, $\alpha = 0.5$, $n_{min} = 50$, tree_depth = 4.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v\log(p_v)$	$b = -p_v^{0.5}\log(p_v)$
Accuracy	Mean	**0.9970**	0.8502	0.8134	0.8852	0.8119	0.7981	0.8253	0.8216	0.8556
	Std	0.0030	0.0997	0.0963	0.0938	0.0931	0.0970	0.0907	0.0989	0.0988
Precision	Mean	**0.9970**	0.7971	0.7778	0.8347	0.7908	0.7905	0.7543	0.7897	0.8002
	Std	0.0029	0.1217	0.1169	0.1294	0.1205	0.1215	0.1255	0.1229	0.1222
Recall	Mean	**0.9970**	0.8502	0.8134	0.8852	0.8119	0.7981	0.8253	0.8216	0.8556
	Std	0.0030	0.0997	0.0963	0.0938	0.0931	0.0970	0.0907	0.0989	0.0988
F1 score	Mean	**0.9970**	0.8060	0.7668	0.8494	0.7660	0.7519	0.7757	0.7759	0.8116
	Std	0.0030	0.1253	0.1112	0.1221	0.1062	0.1083	0.1153	0.1171	0.1257
ARI	Mean	**0.9931**	0.8335	0.7662	0.8799	0.7640	0.7399	0.7923	0.7796	0.8439
	Std	0.0068	0.1158	0.1403	0.0975	0.1353	0.1436	0.1152	0.1396	0.1077

Table 8.1: $N = 2000$, $V = 15$, $K = 8$, $\alpha = 0.85$, $n_{min} = 50$, tree_depth = 3.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v\log(p_v)$	$b = -p_v^{0.5}\log(p_v)$
Accuracy	Mean	**0.6737**	0.6307	0.6105	0.6224	0.5946	0.5654	0.5919	0.6148	0.6399
	Std	0.0221	0.0307	0.0362	0.0423	0.0472	0.0534	0.0468	0.0397	0.0315
Precision	Mean	**0.6241**	0.5987	0.5828	0.5591	0.5701	0.5566	0.5462	0.5872	0.6147
	Std	0.0252	0.0377	0.0452	0.0703	0.0636	0.0694	0.0979	0.0490	0.0428
Recall	Mean	**0.6737**	0.6307	0.6105	0.6224	0.5946	0.5654	0.5919	0.6148	0.6399
	Std	0.0221	0.0307	0.0362	0.0423	0.0472	0.0534	0.0468	0.0397	0.0315
F1 score	Mean	**0.6395**	0.5871	0.5617	0.5686	0.5437	0.5194	0.5148	0.5659	0.5971
	Std	0.0235	0.0359	0.0403	0.0572	0.0564	0.0645	0.0704	0.0454	0.0411
ARI	Mean	**0.4653**	0.4296	0.4119	0.4278	0.3920	0.3497	0.4266	0.4147	0.4317
	Std	0.0316	0.0347	0.0498	0.0404	0.0595	0.0632	0.0285	0.0521	0.0350

Table 8.2: $N = 2000$, $V = 15$, $K = 8$, $\alpha = 0.85$, $n_{min} = 50$, tree_depth = 4.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v\log(p_v)$	$b = -p_v^{0.5}\log(p_v)$
Accuracy	Mean	0.7667	0.7662	0.7462	0.7284	0.7306	0.7039	0.6804	0.7598	**0.7682**
	Std	0.0176	0.0198	0.0369	0.0368	0.0398	0.0503	0.0300	0.0404	0.0208
Precision	Mean	**0.7861**	0.7860	0.7709	0.7508	0.7569	0.7355	0.7126	0.7813	0.7848
	Std	0.0174	0.0186	0.0280	0.0424	0.0363	0.0529	0.0442	0.0324	0.0195
Recall	Mean	0.7667	0.7662	0.7462	0.7284	0.7306	0.7039	0.6804	0.7598	**0.7682**
	Std	0.0176	0.0198	0.0369	0.0368	0.0398	0.0503	0.0300	0.0404	0.0208
F1 score	Mean	0.7686	0.7680	0.7481	0.7205	0.7316	0.7015	0.6536	0.7607	**0.7691**
	Std	0.0175	0.0193	0.0381	0.0472	0.0448	0.0629	0.0366	0.0420	0.0203
ARI	Mean	0.5415	0.5399	0.5150	0.5065	0.4977	0.4616	0.4968	0.5352	**0.5452**
	Std	0.0295	0.0340	0.0487	0.0409	0.0515	0.0553	0.0331	0.0547	0.0361

Table 9.1: $N = 2000$, $V = 6$, $K = 15$, $\alpha = 0.5$, $n_{min} = 50$, tree_depth = 3.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	**0.5115**	0.4570	0.4700	0.4863	0.4624	0.4410	0.5077	0.4694	0.4602
	Std	0.0200	0.0305	0.0305	0.0383	0.0446	0.0559	0.0233	0.0309	0.0350
Precision	Mean	0.3076	0.2902	0.3269	0.3022	0.3374	0.3429	**0.3491**	0.3254	0.2913
	Std	0.0208	0.0355	0.0440	0.0440	0.0448	0.0428	0.0244	0.0421	0.0381
Recall	Mean	**0.5115**	0.4570	0.4700	0.4863	0.4624	0.4410	0.5077	0.4694	0.4602
	Std	0.0200	0.0305	0.0305	0.0383	0.0446	0.0559	0.0233	0.0309	0.0350
F1 score	Mean	0.3705	0.3322	0.3575	0.3546	0.3555	0.3401	**0.3908**	0.3578	0.3328
	Std	0.0215	0.0312	0.0393	0.0437	0.0492	0.0542	0.0246	0.0389	0.0370
ARI	Mean	**0.5668**	0.4315	0.4075	0.4896	0.3801	0.3386	0.4900	0.4038	0.4434
	Std	0.0312	0.0388	0.0345	0.0311	0.0591	0.0799	0.0236	0.0374	0.0389

Table 9.2: $N = 2000$, $V = 6$, $K = 15$, $\alpha = 0.5$, $n_{min} = 50$, tree_depth = 4.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	**0.9004**	0.7271	0.6941	0.8026	0.6805	0.6535	0.7158	0.6920	0.7435
	Std	0.0291	0.0601	0.0388	0.0443	0.0532	0.0675	0.0603	0.0413	0.0585
Precision	Mean	**0.8595**	0.6603	0.6531	0.7310	0.6545	0.6648	0.6242	0.6537	0.6718
	Std	0.0413	0.0730	0.0614	0.0698	0.0665	0.0666	0.0744	0.0640	0.0644
Recall	Mean	**0.9004**	0.7271	0.6941	0.8026	0.6805	0.6535	0.7158	0.6920	0.7435
	Std	0.0291	0.0601	0.0388	0.0443	0.0532	0.0675	0.0603	0.0413	0.0585
F1 score	Mean	**0.8731**	0.6605	0.6272	0.7479	0.6155	0.5935	0.6361	0.6221	0.6785
	Std	0.0381	0.0698	0.0460	0.0572	0.0557	0.0686	0.0782	0.0487	0.0667
ARI	Mean	**0.8580**	0.6716	0.6239	0.7599	0.5892	0.5153	0.7082	0.6270	0.6903
	Std	0.0372	0.0648	0.0518	0.0477	0.0831	0.1131	0.0466	0.0532	0.0672

Table 10.1: $N = 2000$, $V = 6$, $K = 15$, $\alpha = 0.85$, $n_{min} = 50$, tree_depth = 3.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	0.3444	0.3508	**0.3534**	0.3390	0.3477	0.3379	0.3381	0.3527	0.3489
	Std	0.0193	0.0187	0.0179	0.0200	0.0185	0.0194	0.0204	0.0180	0.0188
Precision	Mean	0.1858	0.2074	**0.2107**	0.1812	0.2083	0.2070	0.1776	0.2098	0.2036
	Std	0.0214	0.0178	0.0174	0.0243	0.0179	0.0187	0.0213	0.0167	0.0196
Recall	Mean	0.3444	0.3508	**0.3534**	0.3390	0.3477	0.3379	0.3381	0.3527	0.3489
	Std	0.0193	0.0187	0.0179	0.0200	0.0185	0.0194	0.0204	0.0180	0.0188
F1 score	Mean	0.2328	0.2464	**0.2489**	0.2260	0.2448	0.2385	0.2223	0.2485	0.2429
	Std	0.0193	0.0178	0.0170	0.0212	0.0177	0.0177	0.0191	0.0173	0.0193
ARI	Mean	0.2331	0.2371	0.2372	0.2301	0.2312	0.2116	0.2339	0.2383	**0.2387**
	Std	0.0179	0.0171	0.0167	0.0184	0.0168	0.0227	0.0196	0.0176	0.0173

Table 10.2: $N = 2000$, $V = 6$, $K = 15$, $\alpha = 0.85$, $n_{min} = 50$, tree_depth = 4.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	0.4993	0.5151	0.5116	0.4835	0.5052	0.4783	0.4888	0.5140	**0.5153**
	Std	0.0206	0.0196	0.0190	0.0258	0.0211	0.0335	0.0295	0.0210	0.0198
Precision	Mean	0.4565	0.4842	0.4910	0.4274	0.4882	0.4671	0.4284	**0.4928**	0.4825
	Std	0.0267	0.0245	0.0233	0.0436	0.0249	0.0330	0.0466	0.0241	0.0260
Recall	Mean	0.4993	0.5151	0.5116	0.4835	0.5052	0.4783	0.4888	0.5140	**0.5153**
	Std	0.0206	0.0196	0.0190	0.0258	0.0211	0.0335	0.0295	0.0210	0.0198
F1 score	Mean	0.4615	0.4812	0.4803	0.4330	0.4740	0.4466	0.4369	**0.4825**	0.4801
	Std	0.0225	0.0203	0.0184	0.0335	0.0200	0.0335	0.0377	0.0206	0.0220
ARI	Mean	0.2926	0.2918	0.2852	0.2805	0.2804	0.2632	0.2867	0.2868	**0.2946**
	Std	0.0200	0.0212	0.0208	0.0230	0.0218	0.0272	0.0255	0.0224	0.0204

Table 11.1: $N = 2000$, $V = 15$, $K = 15$, $\alpha = 0.5$, $n_{min} = 50$, tree_depth = 3.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	**0.5082**	0.3900	0.4049	0.3906	0.3807	0.3215	0.3948	0.4012	0.3942
	Std	0.0171	0.0434	0.0567	0.0191	0.0708	0.0678	0.0193	0.0592	0.0438
Precision	Mean	0.2977	0.2950	**0.3126**	0.2933	0.3083	0.3096	0.3018	0.3086	0.2944
	Std	0.0186	0.0406	0.0537	0.0233	0.0580	0.0512	0.0278	0.0509	0.0394
Recall	Mean	**0.5082**	0.3900	0.4049	0.3906	0.3807	0.3215	0.3948	0.4012	0.3942
	Std	0.0171	0.0434	0.0567	0.0191	0.0708	0.0678	0.0193	0.0592	0.0438
F1 score	Mean	**0.3651**	0.3045	0.3227	0.3066	0.2989	0.2570	0.3117	0.3168	0.3085
	Std	0.0189	0.0385	0.0549	0.0182	0.0698	0.0597	0.0213	0.0571	0.0380
ARI	Mean	**0.5639**	0.2347	0.2426	0.1957	0.2269	0.1505	0.2166	0.2471	0.2452
	Std	0.0135	0.0685	0.0792	0.0289	0.0818	0.0775	0.0481	0.0817	0.0738

Table 11.2: $N = 2000$, $V = 15$, $K = 15$, $\alpha = 0.5$, $n_{min} = 50$, tree_depth = 4.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	**0.9023**	0.5426	0.5552	0.5296	0.5362	0.4595	0.5228	0.5557	0.5494
	Std	0.0142	0.0667	0.0793	0.0321	0.0843	0.0891	0.0290	0.0797	0.0728
Precision	Mean	**0.8728**	0.4791	0.5090	0.4541	0.5087	0.4975	0.4379	0.5129	0.4852
	Std	0.0259	0.0699	0.0748	0.0415	0.0761	0.0653	0.0425	0.0673	0.0692
Recall	Mean	**0.9023**	0.5426	0.5552	0.5296	0.5362	0.4595	0.5228	0.5557	0.5494
	Std	0.0142	0.0667	0.0793	0.0321	0.0843	0.0891	0.0290	0.0797	0.0728
F1 score	Mean	**0.8794**	0.4712	0.4903	0.4605	0.4699	0.4023	0.4476	0.4895	0.4779
	Std	0.0190	0.0690	0.0804	0.0375	0.0882	0.0832	0.0376	0.0812	0.0736
ARI	Mean	**0.8768**	0.3928	0.4045	0.3082	0.3656	0.2599	0.3338	0.4040	0.4120
	Std	0.0200	0.0963	0.1129	0.0599	0.1117	0.1135	0.0732	0.1177	0.1016

Table 12.1: $N = 2000$, $V = 15$, $K = 15$, $\alpha = 0.85$, $n_{min} = 50$, tree_depth = 3.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	**0.3972**	0.3825	0.3836	0.3289	0.3826	0.3801	0.3549	0.3817	0.3832
	Std	0.0250	0.0228	0.0197	0.0330	0.0202	0.0238	0.0250	0.0193	0.0227
Precision	Mean	0.2376	0.2710	0.2768	0.1873	0.2769	**0.2816**	0.2387	0.2744	0.2722
	Std	0.0204	0.0255	0.0268	0.0365	0.0249	0.0280	0.0234	0.0267	0.0253
Recall	Mean	**0.3972**	0.3825	0.3836	0.3289	0.3826	0.3801	0.3549	0.3817	0.3832
	Std	0.0250	0.0228	0.0197	0.0330	0.0202	0.0238	0.0250	0.0193	0.0227
F1 score	Mean	0.2883	0.2984	**0.3010**	0.2183	0.3001	0.2991	0.2575	0.2990	0.2991
	Std	0.0222	0.0213	0.0197	0.0361	0.0192	0.0229	0.0261	0.0195	0.0208
ARI	Mean	**0.2612**	0.2066	0.2035	0.2198	0.2024	0.1957	0.2281	0.2034	0.2077
	Std	0.0216	0.0223	0.0243	0.0238	0.0236	0.0207	0.0232	0.0233	0.0255

Table 12.2: $N = 2000$, $V = 15$, $K = 15$, $\alpha = 0.85$, $n_{min} = 50$, tree_depth = 4.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	**0.5573**	0.5150	0.5042	0.4682	0.5032	0.4984	0.4730	0.5088	0.5171
	Std	0.0311	0.0254	0.0249	0.0479	0.0250	0.0268	0.0384	0.0262	0.0289
Precision	Mean	0.5278	0.5298	0.5132	0.4113	0.5157	0.5089	0.4343	0.5195	**0.5364**
	Std	0.0430	0.0327	0.0341	0.0737	0.0360	0.0368	0.0605	0.0369	0.0387
Recall	Mean	**0.5573**	0.5150	0.5042	0.4682	0.5032	0.4984	0.4730	0.5088	0.5171
	Std	0.0311	0.0254	0.0249	0.0479	0.0250	0.0268	0.0384	0.0262	0.0289
F1 score	Mean	**0.5251**	0.4900	0.4778	0.4030	0.4759	0.4704	0.4082	0.4795	0.4914
	Std	0.0363	0.0266	0.0260	0.0592	0.0255	0.0257	0.0507	0.0272	0.0290
ARI	Mean	**0.3485**	0.2821	0.2793	0.2980	0.2772	0.2735	0.3072	0.2819	0.2824
	Std	0.0263	0.0308	0.0304	0.0364	0.0279	0.0288	0.0318	0.0313	0.0346

Real-World UCI Data

Datasets from the UCI Machine Learning Repository used in the experiments.

No.	Dataset	N	V	K
1	Breast Cancer (Prognostic)	194	32	2
2	Balance Scale	625	4	3
3	Car Evaluation	1728	6	4
4	Connectionist Bench Sonar	208	60	2
5	Contraceptive Method Choice	1473	9	3
6	Hayes-Roth	132	4	3
7	Hepatitis	80	19	2
8	Ionosphere	351	34	2
9	Glass	214	9	6
10	Congressional Voting Records	232	16	2
11	Spambase	4601	57	2
12	Mushroom	8214	22	7
13	Yeast	1484	8	10
14	Bach Choral Harmony	5665	15	102
15	Led Display Domain	1000	7	10

Table 1.1. Breast Cancer (Prognostic), tree_depth = 3.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	**0.7110**	0.6955	0.6967	0.6955	0.6959	0.6967	0.6947	0.6959	0.6963
	Std	0.0818	0.0731	0.0726	0.0723	0.0720	0.0720	0.0736	0.0708	0.0725
Precision	Mean	0.6457	0.6558	0.6565	0.6561	0.6563	**0.6572**	0.6559	0.6562	0.6547
	Std	0.0780	0.0874	0.0887	0.0870	0.0869	0.0872	0.0874	0.0866	0.0873
Recall	Mean	**0.7110**	0.6955	0.6967	0.6955	0.6959	0.6967	0.6947	0.6959	0.6963
	Std	0.0818	0.0731	0.0726	0.0723	0.0720	0.0720	0.0736	0.0708	0.0725
F1 score	Mean	0.6657	0.6646	**0.6659**	0.6647	0.6650	0.6658	0.6644	0.6650	0.6648
	Std	0.0736	0.0783	0.0782	0.0778	0.0776	0.0779	0.0784	0.0770	0.0777
ARI	Mean	0.0051	0.0175	**0.0207**	0.0174	0.0176	0.0192	0.0177	0.0170	0.0177
	Std	0.0586	0.0887	0.0913	0.0888	0.0887	0.0869	0.0889	0.0883	0.0899

Table 1.2. Breast Cancer (Prognostic), tree_depth = 4.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	**0.6967**	0.6629	0.6633	0.6625	0.6620	0.6625	0.6600	0.6633	0.6608
	Std	0.0834	0.0639	0.0650	0.0657	0.0658	0.0652	0.0665	0.0634	0.0661
Precision	Mean	**0.6636**	0.6516	0.6528	0.6517	0.6516	0.6520	0.6505	0.6511	0.6508
	Std	0.0947	0.0800	0.0789	0.0780	0.0780	0.0779	0.0783	0.0784	0.0782
Recall	Mean	**0.6967**	0.6629	0.6633	0.6625	0.6620	0.6625	0.6600	0.6633	0.6608
	Std	0.0834	0.0639	0.0650	0.0657	0.0658	0.0652	0.0665	0.0634	0.0661
F1 score	Mean	**0.6676**	0.6499	0.6512	0.6500	0.6497	0.6502	0.6483	0.6500	0.6489
	Std	0.0843	0.0696	0.0699	0.0698	0.0698	0.0695	0.0704	0.0690	0.0701
ARI	Mean	**0.0304**	0.0079	0.0090	0.0073	0.0073	0.0077	0.0061	0.0065	0.0064
	Std	0.0881	0.0650	0.0614	0.0621	0.0621	0.0621	0.0652	0.0623	0.0617

Table 2.1. Balance Scale, tree_depth = 3.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	0.6989	0.7092	0.7055	0.7085	0.7056	0.7046	0.7118	0.7108	**0.7121**
	Std	0.0266	0.0326	0.0319	0.0310	0.0314	0.0317	0.0304	0.0307	0.0351
Precision	Mean	0.6511	0.6612	0.6573	0.6600	0.6571	0.6574	**0.6651**	0.6639	0.6648
	Std	0.0316	0.0370	0.0353	0.0355	0.0342	0.0351	0.0337	0.0335	0.0364
Recall	Mean	0.6989	0.7092	0.7055	0.7085	0.7056	0.7046	0.7118	0.7108	**0.7121**
	Std	0.0266	0.0326	0.0319	0.0310	0.0314	0.0317	0.0304	0.0307	0.0351
F1 score	Mean	0.6707	0.6811	0.6774	0.6804	0.6774	0.6764	0.6835	0.6825	**0.6839**
	Std	0.0292	0.0344	0.0338	0.0326	0.0331	0.0339	0.0316	0.0316	0.0352
ARI	Mean	0.2219	0.2427	0.2361	0.2414	0.2364	0.2339	0.2473	0.2454	**0.2494**
	Std	0.0457	0.0606	0.0581	0.0568	0.0581	0.0574	0.0600	0.0608	0.0713

Table 2.2. Balance Scale, tree_depth = 4.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	0.7442	0.7825	0.7767	0.7782	0.7753	0.7720	0.7761	0.7833	**0.7847**
	Std	0.0261	0.0320	0.0334	0.0310	0.0308	0.0315	0.0276	0.0255	0.0279
Precision	Mean	0.7067	0.7334	0.7331	0.7295	0.7302	0.7288	0.7277	**0.7341**	**0.7341**
	Std	0.0326	0.0357	0.0383	0.0328	0.0391	0.0395	0.0299	0.0316	0.0305
Recall	Mean	0.7442	0.7825	0.7767	0.7782	0.7753	0.7720	0.7761	0.7833	**0.7847**
	Std	0.0261	0.0320	0.0334	0.0310	0.0308	0.0315	0.0276	0.0255	0.0279
F1 score	Mean	0.7199	0.7551	0.7502	0.7508	0.7485	0.7454	0.7487	0.7558	**0.7567**
	Std	0.0280	0.0332	0.0342	0.0310	0.0324	0.0334	0.0277	0.0274	0.0283
ARI	Mean	0.3239	0.4173	0.4031	0.4063	0.3981	0.3900	0.4000	0.4176	**0.4212**
	Std	0.0633	0.0757	0.0773	0.0744	0.0726	0.0745	0.0672	0.0625	0.0697

Table 3.1. Car Evaluation, tree_depth = 3.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	0.7887	0.7878	0.7863	0.7887	0.7896	**0.7969**	0.7887	0.7872	0.7887
	Std	0.0190	0.0190	0.0166	0.0190	0.0150	0.0147	0.0190	0.0181	0.0190
Precision	Mean	0.7300	0.7300	0.7330	0.7300	0.7452	**0.7670**	0.7300	0.7301	0.7300
	Std	0.0271	0.0271	0.0278	0.0271	0.0306	0.0207	0.0271	0.0274	0.0271
Recall	Mean	0.7887	0.7878	0.7863	0.7887	0.7896	**0.7969**	0.7887	0.7872	0.7887
	Std	0.0190	0.0190	0.0166	0.0190	0.0150	0.0147	0.0190	0.0181	0.0190
F1 score	Mean	0.7554	0.7549	0.7554	0.7554	0.7627	**0.7759**	0.7554	0.7545	0.7554
	Std	0.0172	0.0170	0.0164	0.0172	0.0197	0.0164	0.0172	0.0161	0.0172
ARI	Mean	0.5088	0.5061	0.4982	0.5088	0.5008	**0.5109**	0.5088	0.5041	0.5088
	Std	0.0527	0.0530	0.0480	0.0527	0.0389	0.0379	0.0527	0.0532	0.0527

Table 3.2. Car Evaluation, tree_depth = 4.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	0.8339	0.8443	0.8422	0.8443	0.8363	0.8304	0.8443	**0.8444**	0.8443
	Std	0.0183	0.0151	0.0183	0.0151	0.0193	0.0182	0.0151	0.0152	0.0151
Precision	Mean	**0.8523**	0.8451	0.8434	0.8458	0.8385	0.8319	0.8458	0.8445	0.8458
	Std	0.0207	0.0223	0.0214	0.0222	0.0181	0.0144	0.0222	0.0224	0.0222
Recall	Mean	0.8339	0.8443	0.8422	0.8443	0.8363	0.8304	0.8443	**0.8444**	0.8443
	Std	0.0183	0.0151	0.0183	0.0151	0.0193	0.0182	0.0151	0.0152	0.0151
F1 score	Mean	0.8308	0.8371	0.8339	0.8376	0.8254	0.8160	0.8376	0.8367	**0.8376**
	Std	0.0198	0.0186	0.0219	0.0181	0.0224	0.0176	0.0181	0.0188	0.0181
ARI	Mean	0.6506	0.6821	0.6665	**0.6836**	0.6351	0.6047	**0.6836**	0.6794	**0.6836**
	Std	0.0554	0.0489	0.0671	0.0464	0.0720	0.0553	0.0464	0.0534	0.0464

Table 4.1. Connectionist Bench Sonar, tree_depth = 3.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	**0.7211**	0.7162	0.7158	0.7165	0.7135	0.7138	0.7138	0.7158	0.7127
	Std	0.0674	0.0671	0.0659	0.0667	0.0660	0.0661	0.0661	0.0681	0.0651
Precision	Mean	**0.7447**	0.7292	0.7301	0.7300	0.7270	0.7275	0.7274	0.7294	0.7263
	Std	0.0725	0.0703	0.0678	0.0697	0.0681	0.0689	0.0690	0.0707	0.0679
Recall	Mean	**0.7211**	0.7162	0.7158	0.7165	0.7135	0.7138	0.7138	0.7158	0.7127
	Std	0.0674	0.0671	0.0659	0.0667	0.0660	0.0661	0.0661	0.0681	0.0651
F1 score	Mean	**0.7166**	0.7133	0.7125	0.7136	0.7106	0.7108	0.7108	0.7128	0.7097
	Std	0.0690	0.0688	0.0675	0.0685	0.0674	0.0679	0.0679	0.0698	0.0668
ARI	Mean	**0.1978**	0.1892	0.1879	0.1897	0.1839	0.1846	0.1846	0.1891	0.1821
	Std	0.1196	0.1148	0.1128	0.1141	0.1122	0.1125	0.1125	0.1172	0.1101

Table 4.2. Connectionist Bench Sonar, tree_depth = 4.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	**0.7235**	0.7131	0.7135	0.7115	0.7119	0.7108	0.7131	0.7127	0.7119
	Std	0.0696	0.0655	0.0662	0.0658	0.0635	0.0659	0.0647	0.0644	0.0638
Precision	Mean	**0.7366**	0.7229	0.7239	0.7215	0.7216	0.7206	0.7222	0.7228	0.7214
	Std	0.0685	0.0655	0.0650	0.0663	0.0630	0.0657	0.0646	0.0641	0.0632
Recall	Mean	**0.7235**	0.7131	0.7135	0.7115	0.7119	0.7108	0.7131	0.7127	0.7119
	Std	0.0696	0.0655	0.0662	0.0658	0.0635	0.0659	0.0647	0.0644	0.0638
F1 score	Mean	**0.7220**	0.7118	0.7119	0.7103	0.7106	0.7095	0.7118	0.7114	0.7106
	Std	0.0697	0.0667	0.0676	0.0669	0.0648	0.0672	0.0659	0.0657	0.0652
ARI	Mean	**0.2037**	0.1830	0.1840	0.1804	0.1799	0.1792	0.1825	0.1817	0.1801
	Std	0.1232	0.1125	0.1173	0.1142	0.1106	0.1126	0.1112	0.1127	0.1110

Table 5.1. Contraceptive Method Choice, tree_depth = 3.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	0.5016	0.5118	0.5128	0.5131	0.5133	0.5130	**0.5215**	0.5109	0.5143
	Std	0.0293	0.0269	0.0291	0.0279	0.0316	0.0296	0.0246	0.0257	0.0267
Precision	Mean	0.5461	0.5677	0.5374	0.5754	0.5147	0.5043	0.5677	0.5604	**0.5764**
	Std	0.0773	0.0564	0.0677	0.0524	0.0656	0.0734	0.0604	0.0581	0.0510
Recall	Mean	0.5016	0.5118	0.5128	0.5131	0.5133	0.5130	**0.5215**	0.5109	0.5143
	Std	0.0293	0.0269	0.0291	0.0279	0.0316	0.0296	0.0246	0.0257	0.0267
F1 score	Mean	0.4855	0.4992	0.4894	0.5052	0.4886	0.4873	**0.5146**	0.4942	0.5069
	Std	0.0428	0.0317	0.0389	0.0348	0.0428	0.0437	0.0387	0.0343	0.0345
ARI	Mean	0.0605	0.0676	0.0687	0.0698	0.0735	0.0732	**0.0808**	0.0659	0.0709
	Std	0.0247	0.0229	0.0284	0.0244	0.0279	0.0258	0.0223	0.0230	0.0241

Table 5.2. Contraceptive Method Choice, tree_depth = 4.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	0.5431	0.5512	0.5433	0.5497	0.5373	0.5306	0.5492	0.5477	**0.5524**
	Std	0.0258	0.0237	0.0252	0.0227	0.0241	0.0268	0.0223	0.0246	0.0209
Precision	Mean	0.5619	**0.5648**	0.5594	0.5618	0.5600	0.5565	0.5652	0.5625	0.5628
	Std	0.0367	0.0284	0.0291	0.0260	0.0295	0.0401	0.0263	0.0291	0.0262
Recall	Mean	0.5431	0.5512	0.5433	0.5497	0.5373	0.5306	0.5492	0.5477	**0.5524**
	Std	0.0258	0.0237	0.0252	0.0227	0.0241	0.0268	0.0223	0.0246	0.0209
F1 score	Mean	0.5370	0.5455	0.5363	0.5450	0.5306	0.5220	0.5469	0.5419	**0.5472**
	Std	0.0322	0.0244	0.0258	0.0242	0.0238	0.0305	0.0232	0.0250	0.0232
ARI	Mean	0.1020	0.1115	0.1035	0.1111	0.0953	0.0886	0.1115	0.1084	**0.1141**
	Std	0.0298	0.0266	0.0294	0.0260	0.0262	0.0266	0.0259	0.0281	0.0244

Table 6.1. Hayes-Roth, tree_depth = 3.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	**0.5528**	0.5503	0.5370	**0.5528**	0.5364	0.5479	**0.5528**	0.5358	**0.5528**
	Std	0.0602	0.0628	0.0437	0.0602	0.0536	0.0635	0.0602	0.0443	0.0602
Precision	Mean	0.3704	0.3730	0.5486	0.3704	0.5613	**0.5848**	0.3704	0.5169	0.3704
	Std	0.0585	0.0605	0.1309	0.0585	0.1251	0.1242	0.0585	0.1364	0.0585
Recall	Mean	**0.5528**	0.5503	0.5370	**0.5528**	0.5364	0.5479	**0.5528**	0.5358	**0.5528**
	Std	0.0602	0.0628	0.0437	0.0602	0.0536	0.0635	0.0602	0.0443	0.0602
F1 score	Mean	0.4241	0.4243	0.5131	0.4241	0.5189	**0.5337**	0.4241	0.4952	0.4241
	Std	0.0630	0.0630	0.0843	0.0630	0.0886	0.0895	0.0630	0.0843	0.0630
ARI	Mean	**0.4657**	0.4584	0.1953	**0.4657**	0.1766	0.1751	**0.4657**	0.2334	**0.4657**
	Std	0.0698	0.0893	0.1582	0.0698	0.1362	0.1309	0.0698	0.1777	0.0698

Table 6.2. Hayes-Roth, tree_depth = 4.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	**0.6564**	**0.6564**	0.5952	**0.6564**	0.6194	0.6303	**0.6564**	0.6000	**0.6564**
	Std	0.0448	0.0448	0.0907	0.0448	0.0964	0.0913	0.0448	0.0831	0.0448
Precision	Mean	0.6759	0.6759	0.6597	0.6759	0.7019	**0.7028**	0.6759	0.6529	0.6759
	Std	0.0625	0.0625	0.0830	0.0625	0.0920	0.0964	0.0625	0.0763	0.0625
Recall	Mean	**0.6564**	**0.6564**	0.5952	**0.6564**	0.6194	0.6303	**0.6564**	0.6000	**0.6564**
	Std	0.0448	0.0448	0.0907	0.0448	0.0964	0.0913	0.0448	0.0831	0.0448
F1 score	Mean	**0.6551**	**0.6551**	0.5896	**0.6551**	0.6095	0.6200	**0.6551**	0.5979	**0.6551**
	Std	0.0476	0.0476	0.0942	0.0476	0.0967	0.0941	0.0476	0.0872	0.0476
ARI	Mean	**0.3449**	**0.3449**	0.2539	**0.3449**	0.2550	0.2709	**0.3449**	0.2583	**0.3449**
	Std	0.0757	0.0757	0.1180	0.0757	0.1244	0.1087	0.0757	0.1053	0.0757

Table 7.1. Hepatitis, tree_depth = 3.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	0.8150	0.8110	0.8100	0.8080	0.8090	0.8100	0.8130	**0.8170**	0.8090
	Std	0.0743	0.0666	0.0700	0.0681	0.0676	0.0693	0.0699	0.0704	0.0683
Precision	Mean	**0.8427**	0.8166	0.8134	0.8137	0.8152	0.8138	0.8215	0.8245	0.8138
	Std	0.0893	0.0977	0.1014	0.1001	0.0991	0.1002	0.0988	0.0973	0.1002
Recall	Mean	0.8150	0.8110	0.8100	0.8080	0.8090	0.8100	0.8130	**0.8170**	0.8090
	Std	0.0743	0.0666	0.0700	0.0681	0.0676	0.0693	0.0699	0.0704	0.0683
F1 score	Mean	**0.8155**	0.8071	0.8050	0.8038	0.8053	0.8049	0.8098	0.8136	0.8044
	Std	0.0809	0.0783	0.0822	0.0805	0.0794	0.0814	0.0808	0.0798	0.0809
ARI	Mean	**0.2458**	0.1992	0.1947	0.1937	0.1970	0.1941	0.2141	0.2256	0.1938
	Std	0.2302	0.1894	0.1916	0.1905	0.1902	0.1901	0.1930	0.1989	0.1905

Table 7.2. Hepatitis, tree_depth = 4.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	**0.8120**	0.8100	0.8080	0.8060	0.8080	0.8090	0.8060	0.8090	0.8080
	Std	0.0637	0.0686	0.0717	0.0712	0.0703	0.0705	0.0712	0.0705	0.0703
Precision	Mean	**0.8377**	0.8280	0.8242	0.8233	0.8267	0.8252	0.8257	0.8283	0.8252
	Std	0.0864	0.0919	0.0964	0.0961	0.0935	0.0949	0.0961	0.0941	0.0948
Recall	Mean	**0.8120**	0.8100	0.8080	0.8060	0.8080	0.8090	0.8060	0.8090	0.8080
	Std	0.0637	0.0686	0.0717	0.0712	0.0703	0.0705	0.0712	0.0705	0.0703
F1 score	Mean	**0.8131**	0.8129	0.8101	0.8080	0.8109	0.8108	0.8088	0.8118	0.8101
	Std	0.0746	0.0763	0.0807	0.0797	0.0777	0.0797	0.0794	0.0771	0.0791
ARI	Mean	0.2218	0.2294	0.2242	0.2178	0.2260	0.2257	0.2221	**0.2296**	0.2240
	Std	0.2077	0.1911	0.1944	0.1987	0.1926	0.1931	0.1943	0.1920	0.1931

Table 8.1. Ionosphere, tree_depth = 3.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	**0.8973**	0.8866	0.8870	0.8868	0.8866	0.8866	0.8859	0.8866	0.8866
	Std	0.0382	0.0418	0.0408	0.0415	0.0418	0.0418	0.0421	0.0418	0.0418
Precision	Mean	**0.9018**	0.8924	0.8931	0.8927	0.8924	0.8924	0.8917	0.8924	0.8924
	Std	0.0332	0.0368	0.0351	0.0362	0.0368	0.0368	0.0373	0.0368	0.0368
Recall	Mean	**0.8973**	0.8866	0.8870	0.8868	0.8866	0.8866	0.8859	0.8866	0.8866
	Std	0.0382	0.0418	0.0408	0.0415	0.0418	0.0418	0.0421	0.0418	0.0418
F1 score	Mean	**0.8967**	0.8857	0.8861	0.8859	0.8857	0.8857	0.8850	0.8857	0.8857
	Std	0.0405	0.0445	0.0435	0.0442	0.0445	0.0445	0.0448	0.0445	0.0445
ARI	Mean	**0.6293**	0.5963	0.5974	0.5970	0.5963	0.5963	0.5942	0.5963	0.5963
	Std	0.1167	0.1265	0.1242	0.1255	0.1265	0.1265	0.1271	0.1265	0.1265

Table 8.2. Ionosphere, tree_depth = 4.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	0.8789	0.8820	0.8816	0.8820	0.8816	0.8816	0.8811	0.8820	**0.8823**
	Std	0.0387	0.0322	0.0315	0.0318	0.0322	0.0322	0.0321	0.0321	0.0317
Precision	Mean	0.8846	0.8860	0.8857	**0.8862**	0.8854	0.8854	0.8851	0.8858	0.8861
	Std	0.0380	0.0323	0.0313	0.0316	0.0323	0.0323	0.0321	0.0324	0.0319
Recall	Mean	0.8789	0.8820	0.8816	0.8820	0.8816	0.8816	0.8811	0.8820	**0.8823**
	Std	0.0387	0.0322	0.0315	0.0318	0.0322	0.0322	0.0321	0.0321	0.0317
F1 score	Mean	0.8758	0.8795	0.8791	0.8795	0.8791	0.8791	0.8785	0.8796	**0.8798**
	Std	0.0396	0.0335	0.0327	0.0330	0.0334	0.0334	0.0333	0.0333	0.0330
ARI	Mean	0.5676	0.5757	0.5742	0.5756	0.5743	0.5743	0.5728	0.5757	**0.5763**
	Std	0.1178	0.1016	0.1002	0.0998	0.1016	0.1016	0.1013	0.1013	0.1003

Table 9.1. Glass, tree_depth = 3.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	0.6289	0.6533	0.6437	0.5711	0.6148	0.6037	0.5607	0.6585	**0.6596**
	Std	0.0553	0.0512	0.0635	0.1033	0.0539	0.0477	0.1078	0.0581	0.0502
Precision	Mean	**0.6315**	0.6040	0.6239	0.5195	0.6043	0.5913	0.4859	0.6239	0.6000
	Std	0.0672	0.0646	0.0829	0.1792	0.0746	0.0704	0.1897	0.0799	0.0687
Recall	Mean	0.6289	0.6533	0.6437	0.5711	0.6148	0.6037	0.5607	0.6585	**0.6596**
	Std	0.0553	0.0512	0.0635	0.1033	0.0539	0.0477	0.1078	0.0581	0.0502
F1 score	Mean	0.6110	0.6134	0.6180	0.5028	0.5903	0.5771	0.4844	**0.6251**	0.6148
	Std	0.0577	0.0517	0.0677	0.1435	0.0589	0.0513	0.1505	0.0634	0.0547
ARI	Mean	0.2880	0.3154	0.3111	0.3092	0.2808	0.2613	0.3060	0.3273	**0.3302**
	Std	0.0744	0.0749	0.0863	0.0725	0.0723	0.0753	0.0773	0.0809	0.0830

Table 9.2. Glass, tree_depth = 4.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	0.6589	0.6578	0.6544	0.6393	0.6315	0.6270	0.5907	**0.6692**	0.6652
	Std	0.0563	0.0569	0.0692	0.0763	0.0545	0.0567	0.1050	0.0576	0.0641
Precision	Mean	**0.6763**	0.6515	0.6649	0.6286	0.6455	0.6504	0.5599	0.6654	0.6459
	Std	0.0592	0.0790	0.0777	0.1033	0.0577	0.0552	0.1680	0.0732	0.0842
Recall	Mean	0.6589	0.6578	0.6544	0.6393	0.6315	0.6270	0.5907	**0.6692**	0.6652
	Std	0.0563	0.0569	0.0692	0.0763	0.0545	0.0567	0.1050	0.0576	0.0641
F1 score	Mean	0.6498	0.6386	0.6414	0.6138	0.6208	0.6174	0.5373	**0.6529**	0.6425
	Std	0.0587	0.0632	0.0736	0.0926	0.0551	0.0533	0.1437	0.0623	0.0698
ARI	Mean	0.3106	0.3119	0.3043	0.3047	0.2794	0.2718	0.3000	**0.3308**	0.3285
	Std	0.0732	0.0825	0.0954	0.0763	0.0727	0.0674	0.0838	0.0816	0.0830

Table 10.1. Congressional Voting Records, tree_depth = 3.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	**0.9514**	0.9493	0.9486	0.9493	0.9493	0.9493	0.9493	0.9493	0.9493
	Std	0.0269	0.0270	0.0266	0.0270	0.0270	0.0270	0.0270	0.0270	0.0270
Precision	Mean	**0.9541**	0.9524	0.9517	0.9524	0.9524	0.9524	0.9524	0.9524	0.9524
	Std	0.0241	0.0241	0.0237	0.0241	0.0241	0.0241	0.0241	0.0241	0.0241
Recall	Mean	**0.9514**	0.9493	0.9486	0.9493	0.9493	0.9493	0.9493	0.9493	0.9493
	Std	0.0269	0.0270	0.0266	0.0270	0.0270	0.0270	0.0270	0.0270	0.0270
F1 score	Mean	**0.9513**	0.9493	0.9486	0.9493	0.9493	0.9493	0.9493	0.9493	0.9493
	Std	0.0270	0.0271	0.0267	0.0271	0.0271	0.0271	0.0271	0.0271	0.0271
ARI	Mean	**0.8146**	0.8070	0.8044	0.8070	0.8070	0.8070	0.8070	0.8070	0.8070
	Std	0.0961	0.0965	0.0948	0.0965	0.0965	0.0965	0.0965	0.0965	0.0965

Table 10.2. Congressional Voting Records, tree_depth = 4.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	0.9490	0.9483	**0.9493**	**0.9493**	0.9486	0.9486	0.9486	0.9483	0.9483
	Std	0.0269	0.0272	0.0277	0.0272	0.0275	0.0275	0.0275	0.0272	0.0272
Precision	Mean	0.9520	0.9513	**0.9523**	0.9521	0.9517	0.9517	0.9517	0.9513	0.9513
	Std	0.0231	0.0241	0.0247	0.0243	0.0244	0.0244	0.0244	0.0241	0.0241
Recall	Mean	0.9490	0.9483	**0.9493**	**0.9493**	0.9486	0.9486	0.9486	0.9483	0.9483
	Std	0.0269	0.0272	0.0277	0.0272	0.0275	0.0275	0.0275	0.0272	0.0272
F1 score	Mean	0.9490	0.9483	**0.9493**	**0.9493**	0.9486	0.9486	0.9486	0.9483	0.9483
	Std	0.0270	0.0272	0.0277	0.0273	0.0275	0.0275	0.0275	0.0272	0.0272
ARI	Mean	0.8058	0.8033	**0.8072**	0.8071	0.8046	0.8046	0.8046	0.8033	0.8033
	Std	0.0960	0.0969	0.0990	0.0975	0.0982	0.0982	0.0982	0.0969	0.0969

Table 11.1. Spambase, tree_depth = 3.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	0.8693	**0.8820**	**0.8820**	**0.8820**	**0.8820**	**0.8820**	**0.8820**	**0.8820**	**0.8820**
	Std	0.0085	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111
Precision	Mean	0.8757	**0.8832**	**0.8832**	**0.8832**	**0.8832**	**0.8832**	**0.8832**	**0.8832**	**0.8832**
	Std	0.0084	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105
Recall	Mean	0.8693	**0.8820**	**0.8820**	**0.8820**	**0.8820**	**0.8820**	**0.8820**	**0.8820**	**0.8820**
	Std	0.0085	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111
F1 score	Mean	0.8667	**0.8810**	**0.8810**	**0.8810**	**0.8810**	**0.8810**	**0.8810**	**0.8810**	**0.8810**
	Std	0.0094	0.0113	0.0113	0.0113	0.0113	0.0113	0.0113	0.0113	0.0113
ARI	Mean	0.5430	**0.5824**	**0.5824**	**0.5824**	**0.5824**	**0.5824**	**0.5824**	**0.5824**	**0.5824**
	Std	0.0258	0.0337	0.0337	0.0337	0.0337	0.0337	0.0337	0.0337	0.0337

Table 11.2. Spambase, tree_depth = 4.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	**0.8982**	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976
	Std	0.0086	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104
Precision	Mean	**0.8997**	0.8995	0.8995	0.8995	0.8995	0.8995	0.8995	0.8996	0.8996
	Std	0.0084	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105
Recall	Mean	**0.8982**	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976
	Std	0.0086	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104
F1 score	Mean	**0.8969**	0.8964	0.8964	0.8964	0.8964	0.8964	0.8964	0.8964	0.8964
	Std	0.0087	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104
ARI	Mean	**0.6326**	0.6310	0.6310	0.6309	0.6310	0.6310	0.6310	0.6310	0.6310
	Std	0.0272	0.0330	0.0330	0.0330	0.0330	0.0330	0.0330	0.0330	0.0330

Table 12.1. Mushroom, tree_depth = 3.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	0.6069	0.6008	0.6012	0.5830	**0.6181**	**0.6181**	0.5807	0.6003	0.5970
	Std	0.0130	0.0112	0.0110	0.0133	0.0098	0.0098	0.0093	0.0110	0.0133
Precision	Mean	**0.5392**	0.5078	0.5063	0.5275	0.4701	0.4701	0.5154	0.5177	0.5334
	Std	0.0418	0.0635	0.0658	0.0490	0.0124	0.0124	0.0344	0.0690	0.0627
Recall	Mean	0.6069	0.6008	0.6012	0.5830	**0.6181**	**0.6181**	0.5807	0.6003	0.5970
	Std	0.0130	0.0112	0.0110	0.0133	0.0098	0.0098	0.0093	0.0110	0.0133
F1 score	Mean	0.5166	0.5262	0.5254	0.5106	0.5240	0.5240	0.5053	**0.5292**	0.5177
	Std	0.0213	0.0221	0.0229	0.0230	0.0117	0.0117	0.0237	0.0242	0.0245
ARI	Mean	0.2979	0.3096	0.3232	0.2755	**0.3593**	**0.3593**	0.2476	0.3076	0.2894
	Std	0.0198	0.0170	0.0198	0.0306	0.0120	0.0120	0.0277	0.0168	0.0206

Table 12.2. Mushroom, tree_depth = 4.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	0.6403	0.6373	0.6362	0.6078	0.6377	**0.6455**	0.6111	0.6368	0.6352
	Std	0.0103	0.0123	0.0119	0.0171	0.0093	0.0101	0.0114	0.0124	0.0174
Precision	Mean	**0.6432**	0.5943	0.6030	0.6077	0.5951	0.5918	0.6431	0.5950	0.6332
	Std	0.0591	0.0415	0.0410	0.0501	0.0702	0.0694	0.0635	0.0395	0.0465
Recall	Mean	0.6403	0.6373	0.6362	0.6078	0.6377	**0.6455**	0.6111	0.6368	0.6352
	Std	0.0103	0.0123	0.0119	0.0171	0.0093	0.0101	0.0114	0.0124	0.0174
F1 score	Mean	0.5940	0.5995	**0.6020**	0.5418	0.5802	0.5846	0.5671	0.6000	0.5863
	Std	0.0298	0.0271	0.0269	0.0264	0.0268	0.0278	0.0310	0.0260	0.0340
ARI	Mean	0.3921	**0.3995**	0.3944	0.3254	0.3565	0.3712	0.3142	0.3991	0.3781
	Std	0.0290	0.0249	0.0257	0.0386	0.0120	0.0129	0.0251	0.0249	0.0311

Table 13.1. Yeast, tree_depth = 3.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	0.5035	0.5598	0.5546	0.4481	0.5470	0.5397	0.4777	0.5591	**0.5652**
	Std	0.0268	0.0204	0.0200	0.0246	0.0185	0.0249	0.0329	0.0195	0.0219
Precision	Mean	0.3637	0.5488	0.5465	0.2620	0.5343	0.5271	0.3249	0.5494	**0.5516**
	Std	0.0440	0.0250	0.0235	0.0620	0.0204	0.0296	0.0568	0.0238	0.0242
Recall	Mean	0.5035	0.5598	0.5546	0.4481	0.5470	0.5397	0.4777	0.5591	**0.5652**
	Std	0.0268	0.0204	0.0200	0.0246	0.0185	0.0249	0.0329	0.0195	0.0219
F1 score	Mean	0.4093	0.5344	0.5295	0.3042	0.5243	0.5174	0.3627	0.5335	**0.5401**
	Std	0.0358	0.0232	0.0219	0.0262	0.0199	0.0266	0.0456	0.0226	0.0249
ARI	Mean	**0.2700**	0.2148	0.2106	0.2259	0.2043	0.2005	0.2557	0.2146	0.2199
	Std	0.0260	0.0214	0.0203	0.0217	0.0177	0.0195	0.0274	0.0204	0.0215

Table 13.2. Yeast, tree_depth = 4.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	**0.5690**	0.5650	0.5632	0.4720	0.5590	0.5568	0.5099	0.5607	0.5633
	Std	0.0207	0.0204	0.0187	0.0292	0.0190	0.0231	0.0273	0.0187	0.0225
Precision	Mean	**0.5631**	0.5653	0.5626	0.3619	0.5569	0.5545	0.3883	0.5549	0.5582
	Std	0.0224	0.0247	0.0207	0.0709	0.0209	0.0260	0.0401	0.0230	0.0257
Recall	Mean	**0.5690**	0.5650	0.5632	0.4720	0.5590	0.5568	0.5099	0.5607	0.5633
	Std	0.0207	0.0204	0.0187	0.0292	0.0190	0.0231	0.0273	0.0187	0.0225
F1 score	Mean	**0.5520**	0.5461	0.5437	0.3485	0.5391	0.5364	0.4169	0.5401	0.5438
	Std	0.0231	0.0227	0.0214	0.0410	0.0215	0.0252	0.0281	0.0208	0.0258
ARI	Mean	0.2445	0.2365	0.2340	0.2424	0.2303	0.2280	**0.2857**	0.2285	0.2351
	Std	0.0224	0.0226	0.0203	0.0260	0.0197	0.0235	0.0242	0.0218	0.0225

Table 14.1. Bach Choral Harmony, tree_depth = 3.

Metric	Statistic	Entropy	$b = 1$	$b = p_v^{0.5}$	$b = \sqrt{p_v(1-p_v)}$	$b = p_v$	$b = p_v^2$	$b = -\log(p_v)$	$b = -p_v \log(p_v)$	$b = -p_v^{0.5} \log(p_v)$
Accuracy	Mean	0.4086	**0.4162**	0.4154	0.2711	0.4108	0.3967	0.3528	0.4154	0.4161
	Std	0.0233	0.0115	0.0145	0.0364	0.0176	0.0187	0.0200	0.0148	0.0109
Precision	Mean	0.2463	0.2659	0.2694	0.1152	0.2655	0.2493	0.1823	**0.2705**	0.2650
	Std	0.0211	0.0132	0.0166	0.0272	0.0200	0.0237	0.0229	0.0185	0.0140
Recall	Mean	0.4086	**0.4162**	0.4154	0.2711	0.4108	0.3967	0.3528	0.4154	0.4161
	Std	0.0233	0.0115	0.0145	0.0364	0.0176	0.0187	0.0200	0.0148	0.0109
F1 score	Mean	0.2930	0.3098	0.3123	0.1486	0.3078	0.2917	0.2302	**0.3130**	0.3091
	Std	0.0236	0.0122	0.0155	0.0361	0.0191	0.0218	0.0217	0.0170	0.0121
ARI	Mean	**0.2690**	0.2675	0.2545	0.1327	0.2532	0.2518	0.2083	0.2516	0.2678
	Std	0.0246	0.0163	0.0164	0.0306	0.0179	0.0178	0.0194	0.0161	0.0186

Table 14.2. Bach Choral Harmony, tree_depth = 4.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	0.5416	0.5422	0.5267	0.3752	0.5210	0.5208	0.4617	0.5256	**0.5456**
	Std	0.0186	0.0121	0.0175	0.0574	0.0184	0.0187	0.0221	0.0167	0.0131
Precision	Mean	0.4174	**0.4388**	0.4273	0.2226	0.4211	0.4144	0.3095	0.4288	0.4359
	Std	0.0261	0.0186	0.0319	0.0624	0.0287	0.0271	0.0303	0.0355	0.0190
Recall	Mean	0.5416	0.5422	0.5267	0.3752	0.5210	0.5208	0.4617	0.5256	**0.5456**
	Std	0.0186	0.0121	0.0175	0.0574	0.0184	0.0187	0.0221	0.0167	0.0131
F1 score	Mean	0.4586	0.4687	0.4553	0.2608	0.4489	0.4455	0.3572	0.4556	**0.4693**
	Std	0.0239	0.0142	0.0233	0.0643	0.0225	0.0210	0.0269	0.0243	0.0149
ARI	Mean	0.3979	**0.4051**	0.3774	0.2183	0.3643	0.3689	0.3058	0.3717	0.4025
	Std	0.0252	0.0181	0.0282	0.0566	0.0379	0.0316	0.0277	0.0300	0.0243

Table 15.1. LED Display Domain, tree_depth = 3.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	**0.5616**	0.5324	0.5152	0.5478	0.4764	0.4384	0.5418	0.5109	0.5379
	Std	0.0399	0.0357	0.0467	0.0468	0.0624	0.0579	0.0533	0.0568	0.0398
Precision	Mean	**0.4861**	0.4543	0.4316	0.4717	0.3888	0.3519	0.4618	0.4275	0.4629
	Std	0.0489	0.0518	0.0606	0.0611	0.0745	0.0706	0.0718	0.0749	0.0536
Recall	Mean	**0.5616**	0.5324	0.5152	0.5478	0.4764	0.4384	0.5418	0.5109	0.5379
	Std	0.0399	0.0357	0.0467	0.0468	0.0624	0.0579	0.0533	0.0568	0.0398
F1 score	Mean	**0.5082**	0.4749	0.4518	0.4928	0.4068	0.3635	0.4849	0.4477	0.4821
	Std	0.0465	0.0453	0.0552	0.0556	0.0733	0.0697	0.0649	0.0690	0.0481
ARI	Mean	**0.3782**	0.3494	0.3456	0.3609	0.3130	0.2767	0.3578	0.3380	0.3527
	Std	0.0403	0.0353	0.0448	0.0471	0.0554	0.0434	0.0532	0.0491	0.0385

Table 15.2. LED Display Domain, tree_depth = 4.

Metric	Statistic	Entropy	$b=1$	$b=p_v^{0.5}$	$b=\sqrt{p_v(1-p_v)}$	$b=p_v$	$b=p_v^2$	$b=-\log(p_v)$	$b=-p_v\log(p_v)$	$b=-p_v^{0.5}\log(p_v)$
Accuracy	Mean	**0.6999**	0.6735	0.6673	0.6711	0.6330	0.5946	0.6714	0.6602	0.6735
	Std	0.0290	0.0404	0.0518	0.0452	0.0624	0.0580	0.0475	0.0525	0.0389
Precision	Mean	**0.7247**	0.6937	0.6860	0.6920	0.6448	0.5998	0.6870	0.6793	0.6930
	Std	0.0361	0.0510	0.0672	0.0551	0.0873	0.0929	0.0611	0.0680	0.0506
Recall	Mean	**0.6999**	0.6735	0.6673	0.6711	0.6330	0.5946	0.6714	0.6602	0.6735
	Std	0.0290	0.0404	0.0518	0.0452	0.0624	0.0580	0.0475	0.0525	0.0389
F1 score	Mean	**0.7022**	0.6708	0.6625	0.6680	0.6158	0.5626	0.6663	0.6531	0.6704
	Std	0.0323	0.0473	0.0647	0.0519	0.0823	0.0770	0.0564	0.0666	0.0460
ARI	Mean	**0.4485**	0.4299	0.4271	0.4287	0.4076	0.3894	0.4332	0.4218	0.4307
	Std	0.0379	0.0430	0.0485	0.0454	0.0479	0.0464	0.0446	0.0463	0.0408